Semidiscrete models for gravitational fingering in the TFE regime

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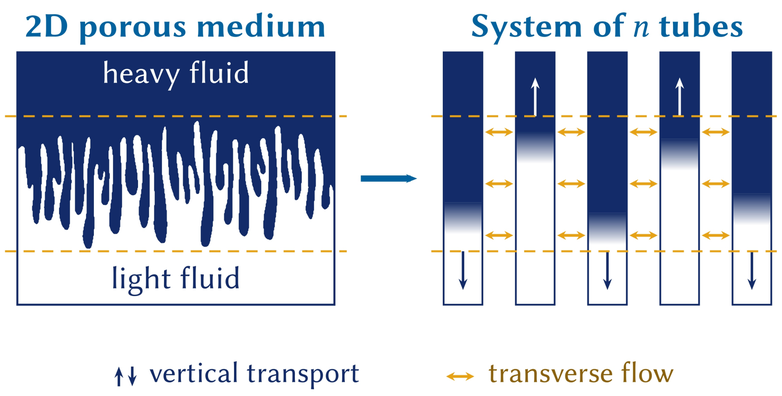
Motivation

Consider a porous medium in which a heavier fluid lies above a lighter, miscible one.

The interface is unstable: small perturbations grow into fingers. This phenomenon is called *gravitational fingering*.

The region where the two fluids interact is the **mixing zone**, and the goal of this study is to understand how fast it grows.

To this end, we replace the continuous medium by a system of n interconnected vertical tubes, one for each "finger". This is the **semidiscrete model**.



Model

In a system of n tubes, the evolution of the fluid concentration in the k-th tube, $c_k(t, y)$, is formulated as follows:

$$\underbrace{\partial_t c_k + \partial_y(u_k c_k)}_{\text{transport}} - \underbrace{\partial_{yy} c_k}_{\text{diffusion}} = \underbrace{F_k(c_1, \dots, c_n)}_{\text{transverse flow}}$$

where the fluid velocity in the k-th tube, u_k, is determined by the TFE regime (*Transverse Flow Equilibrium*):

$$u_k = \left(\frac{1}{n}\sum_{i=1}^{n} c_i\right) - c_k := \bar{c} - c_k$$

We adopt the *Lax–Friedrichs* discretization of the transverse flow function F_k:

$$F_k(c_1, \dots, c_n) := -\frac{1}{n}\sum_{j=1}^{n} w_{kj}\frac{c_j + c_k}{2}$$

In this project, we consider the two-tube case (n=2). Substituting these definitions into the concentration equation, we obtain the following nonlinear PDE:

$$\partial_t c + A(c)\partial_y c = \partial_{yy} c,$$
$$c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \quad A(c) = \frac{(c_1 - c_2)}{4}\begin{pmatrix} -3 & 1 \\ -1 & 3 \end{pmatrix}. \qquad \text{(I)}$$

Diagonalizing the constant matrix above reveals a convenient change of variables:

$$g_1 = \frac{c_1 - k c_2}{k-1}, \quad g_2 = \frac{k c_1 - c_2}{k-1}, \quad k = 3 + 2\sqrt{2}.$$

Rescaling $y \mapsto y/2$ and $t \mapsto t/4$, the system becomes:

$$\begin{cases} \partial_t g_1 + (g_1 + g_2)\partial_y g_1 = \partial_{yy} g_1 \\ \partial_t g_2 - (g_1 + g_2)\partial_y g_2 = \partial_{yy} g_2 \end{cases} \qquad \text{(II)}$$

Thus, the problem reduces to a **system of coupled Burgers equations.**

Definitions

I. Traveling wave:

$f(t, y) : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}^n$, $n \in \mathbb{N}$, such that $f(t, y) = W(y - vt)$ is continuous, with:

- $v \in \mathbb{R}$
- $W(-\infty) = f_- \in \mathbb{R}^n$ and $W(+\infty) = f_+ \in \mathbb{R}^n$

Then f is a traveling wave with speed v from f_- to f_+. We call W the wave profile.

$$f_- \xrightarrow{W} f_+$$

II. Propagating terrace: *(of two traveling waves)*

$\{c_-, c_0, c_+\} \subset \mathbb{R}^n$ and $v_1, v_2 \in \mathbb{R}$, such that there exist wave profiles $W_1(y - v_1 t)$ and $W_2(y - v_2 t)$ satisfying:

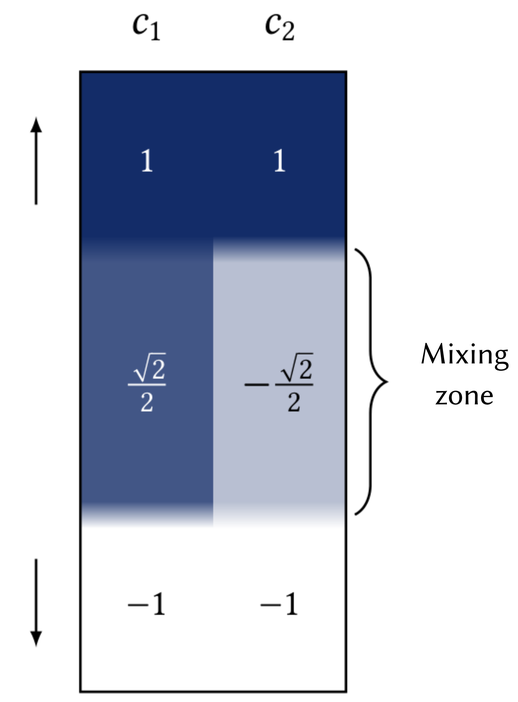
- $W_1(-\infty) = c_-$
- $W_1(+\infty) = W_2(-\infty) = c_0$
- $W_2(+\infty) = c_+$
- $v_1 < v_2$

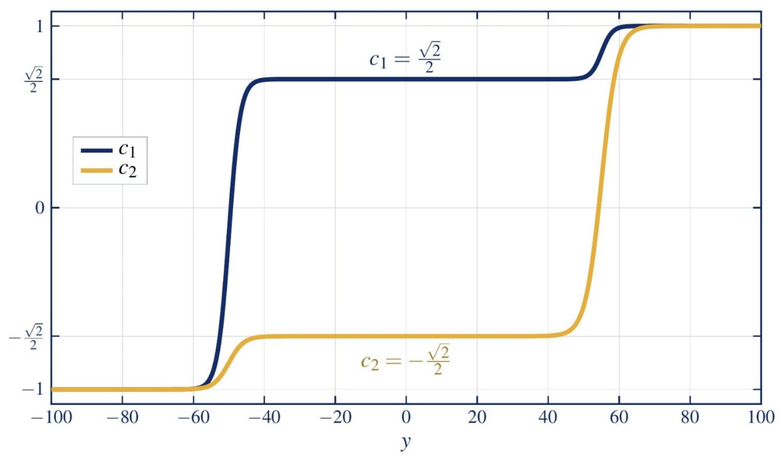
$$c_- \xrightarrow{W_1} c_0 \xrightarrow{W_2} c_+$$

Main Result

Theorem. *For system (I), there exist exactly two propagating terraces of two traveling waves connecting the states* $(-1, -1)$ *and* $(1, 1)$.

Moreover, the waves move with the constant speeds $v_1 = -\frac{1}{2}$ *and* $v_2 = \frac{1}{2}$, *and the terraces pass through the intermediate equilibrium states* $(c_1, c_2) = \pm\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.





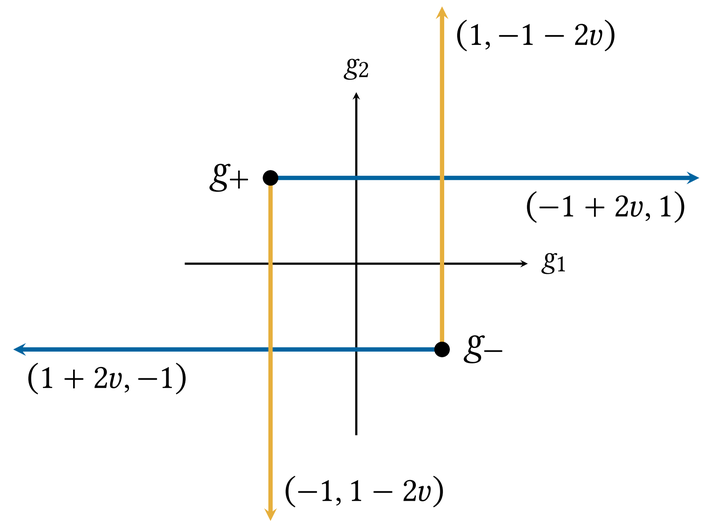
Idea of the Proof

I. Existence: Looking for traveling waves $g(\xi) = g(y - vt)$ reduces system (II) to the ODEs:

$$\begin{cases} \dot{g}_1 = h_1 \\ \dot{g}_2 = h_2 \end{cases} \quad \text{and} \quad \begin{cases} \dot{h}_1 = h_1(g_1 + g_2 - v) \\ \dot{h}_2 = -h_2(g_1 + g_2 + v) \end{cases}$$

The states $c_- = (-1, -1)$ and $c_+ = (1, 1)$ correspond to $g_- = (1, -1)$ and $g_+ = (-1, 1)$. We analyze the invariant planes $h_1 = 0$ and $h_2 = 0$.

Integrating the system along these planes, we parametrize the trajectories in terms of v, which allows us to determine exactly the intermediate states where the waves connect.



II. Uniqueness: We analyze the derivative of the expression $P(\xi) = h_1(\xi)\, h_2(\xi)$:

$$\dot{P} = \dot{h}_1 h_2 + h_1 \dot{h}_2 = h_1 h_2(-2v) = -2vP$$

The analytical solution of this ODE is $P(\xi) = P_0 e^{-2v\xi}$. Since $h_1, h_2 \to 0$ as $\xi \to \pm\infty$, the function $P(\xi)$ must be bounded. However, if $P_0 \neq 0$, the exponential diverges at one of the infinities. Hence:

$$P_0 = 0 \implies h_1(\xi)\, h_2(\xi) \equiv 0$$

We conclude that all connections must be contained in the planes $h_1 = 0$ or $h_2 = 0$, which guarantees the uniqueness of the propagating terraces.

Conclusion

Model	Wave speeds	Intermediate states	Mixing zone growth
Upwind [1]	$v_1 = -\frac{1}{4}, v_2 = \frac{1}{4}$	$\pm\left(\frac{1}{2}, -\frac{1}{2}\right)$	$(v_2 - v_1)t = \frac{1}{2}t$
Lax–Friedrichs	$v_1 = -\frac{1}{2}, v_2 = \frac{1}{2}$	$\pm\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$	$(v_2 - v_1)t = t$

References and More

[1] Petrova, Y.; Tikhomirov, S.; Efendiev, Y. Propagating Terrace in a Two-Tubes Model of Gravitational Fingering. *SIAM Journal on Mathematical Analysis*, v. 57, n. 1, p. 30-64, 2025.

[2] Robinson, C. *An Introduction to Dynamical Systems: Continuous and Discrete*. 2. ed. Providence: American Mathematical Society, 2012.

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